

Well Hydraulics

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Presentation Outline

1. Fundamentals of Well Hydraulics
2. Steady Radial Flow to Non-Leaky Wells
3. Unsteady Radial Flow to Non-Leaky Wells
4. Partially penetrating wells
5. Multiple Well Systems and Interference

Fundamentals of Well Hydraulics

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What is Well Hydraulics?

- A branch of **hydrogeology** concerned with the **movement of groundwater** and **pressure behavior** in wells. (Testbook, n.d.)
- Examines how groundwater flows through aquifers toward wells and how pumping affects water levels.
- Plays a critical role in:
 - I. Groundwater resource management
 - II. Well design and optimization
 - III. Assessment of well performance

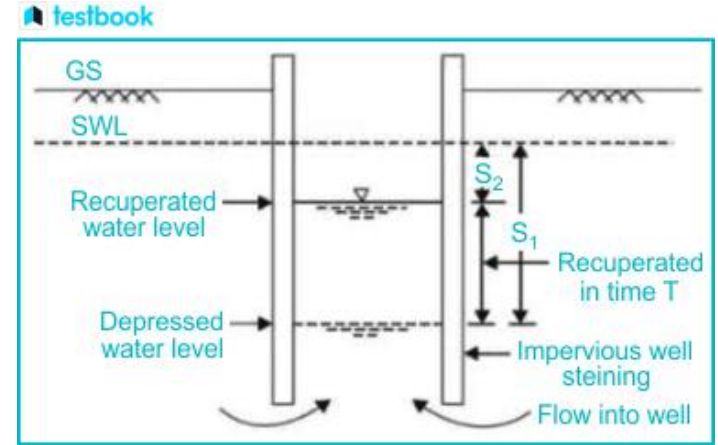


Fig: Recuperating Test in Well Hydraulics
Source: Testbook (n.d.)

Aquifers and Groundwater Flow

- ❖ **Aquifer:** Permeable geological formation that stores and transmits groundwater.
- ❖ Groundwater moves through pore spaces and fractures.
- ❖ Flow depends on:
 - Hydraulic gradient
 - Permeability of the formation
 - Aquifer thickness
- ❖ Darcy's law: $Q = -kA \frac{dh}{dL}$

Aquifers and wells

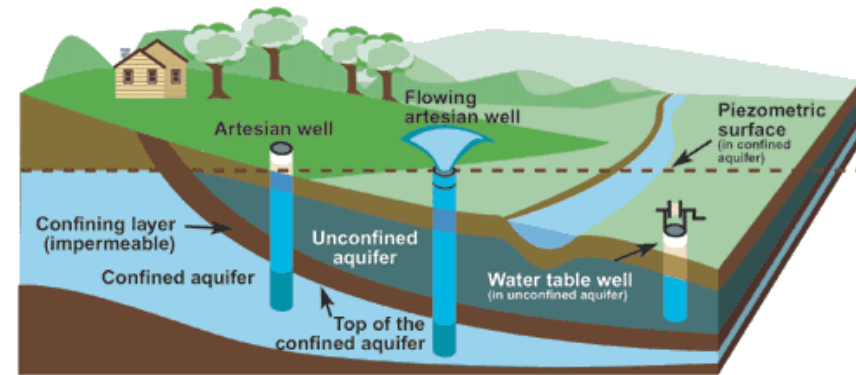


Fig: Artesian well and a flowing artesian well
Source: USGS(2019); credited to Environment and Climate Change Canada

Flow Towards a Pumping Well

- Groundwater occurs in **saturated soil and rock below the water table**.
- Pumping **lowers hydraulic head at the well**.
- This creates a **pressure imbalance** within the aquifer.
- Groundwater flows from surrounding regions **toward the well**.
- The well behaves as a **hydraulic sink**, attracting flow from all directions.
- If pumping exceeds recharge:
 - A **cone of depression** forms around the well
 - Water levels decline most near the well
 - Prolonged pumping can lower the regional

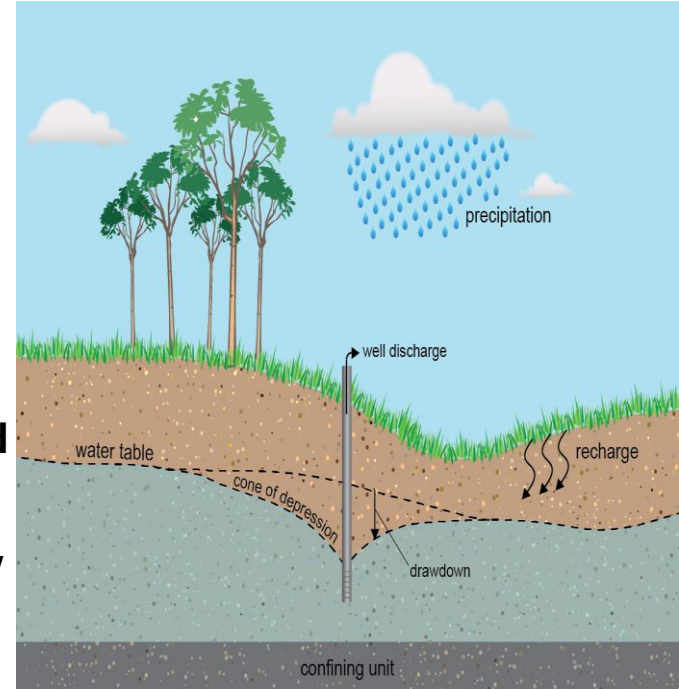


Fig: Cone of depression around the well
Source: USGS(2019); credited to Tara Gross

Natural Flow vs Pumping-Induced Flow

Natural Groundwater Flow

- Controlled by regional hydraulic gradients
- Governed by topography and recharge–discharge zones
- Streamlines are smooth and gently sloping

Pumping-Induced Flow

- Pumping lowers hydraulic head at the well
- Distorts the natural flow field
- Flow lines bend and converge toward the well
- Local gradients dominate, producing radial flow toward a hydraulic sink

(SERC, 2017)

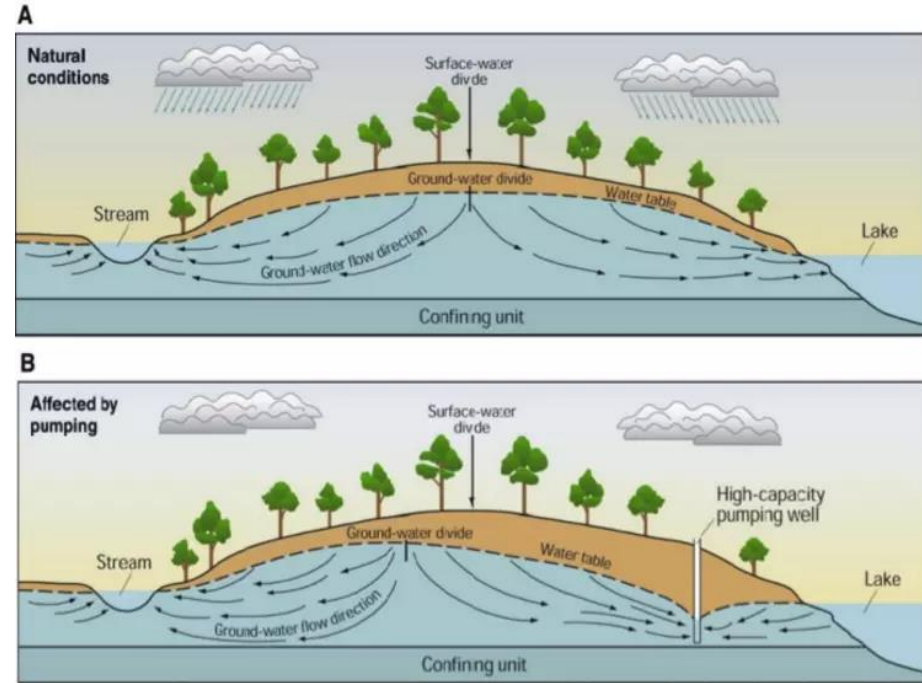


Fig: Effects of pumping and generation of a cone of depression in an unconfined aquifer between a lake and a stream.

Source: USGS: Ground water in the Great Lakes Basin: the case of southeastern Wisconsin

Why Does Flow Become Radial?

- Radial flow develops in an aquifer when water is vertically extracted from or injected into a well. (Yeh & Chang, 2012)
- Wells withdraw water at a localized point.
- In homogeneous and isotropic aquifers: properties are uniform in all directions and every point.
- This symmetry results in:
 - Radial flow toward the well
 - Concentric equipotential surfaces

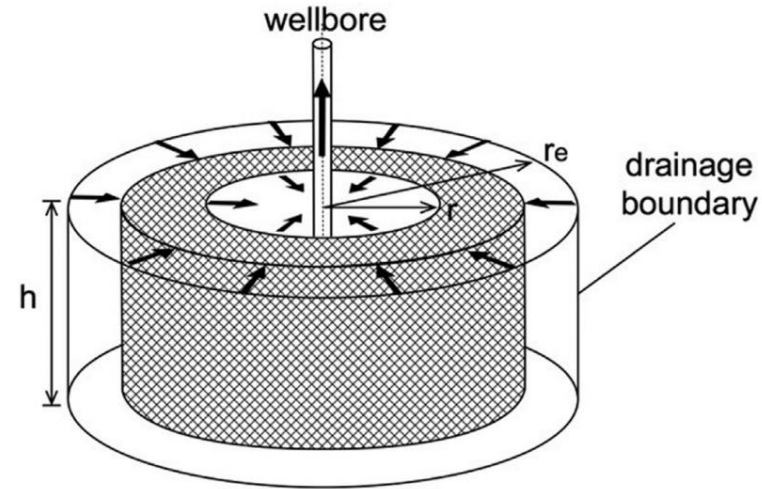


Fig: Wellbore in the center of a cylindrical domain

Source: Balhoff, 2022

Radial Flow Concept

- Preferable in axisymmetric groundwater flow to wells. (Todd and Mays, 2005)
- Flow converges toward the well from all directions, similar to spokes converging at a wheel hub (Fetter, 2000)
- If the aquifer is homogeneous and isotropic in the horizontal plane: flow exhibits radial symmetry; hydraulic head depends only on radial distance (r) and time (t); angular variation (θ) is irrelevant

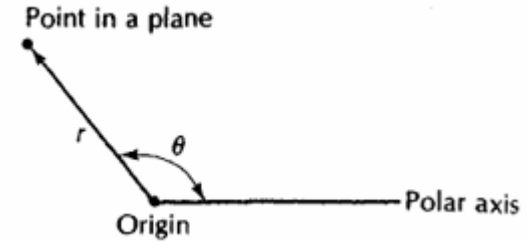


Fig: Polar coordinates
Source: Fetter, 2000

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = \frac{S}{T} \frac{\partial h}{\partial t}$$

$$r = \sqrt{x^2 + y^2}$$

$$\frac{\partial^2 h}{\partial r^2} + \frac{1}{r} \frac{\partial h}{\partial r} = \frac{S}{T} \frac{\partial h}{\partial t}$$

h = hydraulic head (L)

r = radial distance from pumping well (L)

t = time (T)

T = transmissivity (L^2/T)

S = storativity (dimensionless)

Important Terms in Well Hydraulics: Water Levels

Static Water Level (SWL)

- Water level in a well at **equilibrium** (no pumping)
- Measured as **depth to water** from ground surface
- In artesian wells, level may rise **above ground**
- Represents total **hydraulic head**
(water table in unconfined aquifers;
potentiometric surface in confined aquifers)

Pumping Water Level (PWL)

- Water level **during pumping**
- Also called **dynamic water level**
- Lower than SWL due to **drawdown**
- Reflects aquifer and well response to pumping

(Western Oregon University, n.d.;)

Important Terms in Well Hydraulics: Drawdown

- Drawdown (DD) is the **lowering** of water level in a well due to pumping.
- Defined as the difference between **Static Water Level (SWL)** and **Pumping Water Level (PWL)**.
- Indicates the response of an aquifer to pumping and thus used to assess **well performance** and **aquifer characteristics**.
- Higher pumping rates tends to greater drawdown.
- Represents the hydraulic head difference driving groundwater toward the well
- **$DD(s) = SWL - PWL$**
- **Unconfined aquifer:** Drawdown lowers the water table and reduces saturated thickness, whereas in a **confined aquifer** drawdown reflects pressure head reduction without changing saturated thickness, controlled by aquifer properties and pumping.

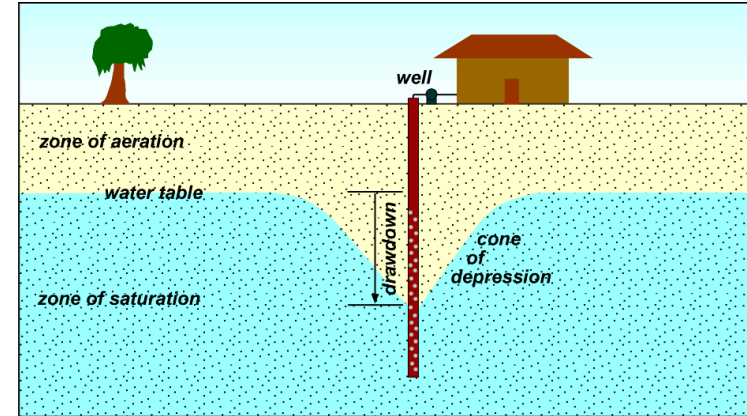


Fig: Drawdown and Cone of Depression in Wells

Source: Groundwater Basics (n.d.), by Columbia University

Important Terms in Well Hydraulics: Cone of Depression

- Pumping a well causes groundwater to flow toward the well, lowering the surrounding water level.
- This lowering forms a cone-shaped surface known as the cone of depression.
- The lowest point of the cone occurs at the pumping well.
- Size and extent depend on: Pumping rate, Aquifer transmissivity & hydraulic conductivity, Storage properties, Duration of pumping.
- **Unconfined aquifers:** cone usually extends < 100 m, but may reach several hundred meters.
- **Confined aquifers:** cone may extend hundreds to thousands of meters due to pressure reduction.
(Cornell University, 2010)

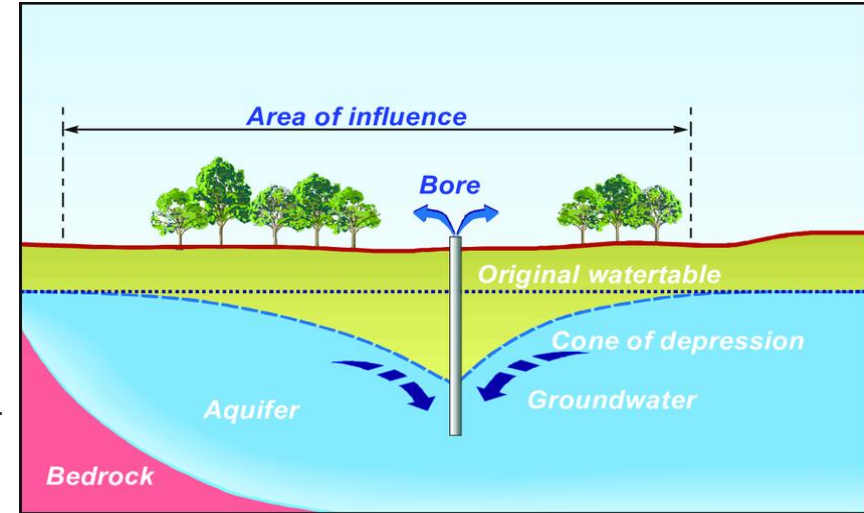


Fig: Cone of Depression
Source: Dogramaci, 2004

Important Terms in Well Hydraulics: Radius of Influence

- The **radius of influence (R)** is the distance from the well center to the edge of the cone of depression.
- Represents the area of influence of a pumping well.
- In unconfined aquifers: actual lowering of water table.
- In confined aquifers: reduction in pressure head, not physical water level.
- Important for: Well spacing design, Avoiding interference between adjacent wells
- Estimated using analytical equations (e.g., Bear, 1979).
- Depends on: Transmissivity (T), Storage coefficient (S), Pumping duration

(Shahinuzzaman et al., 2020)

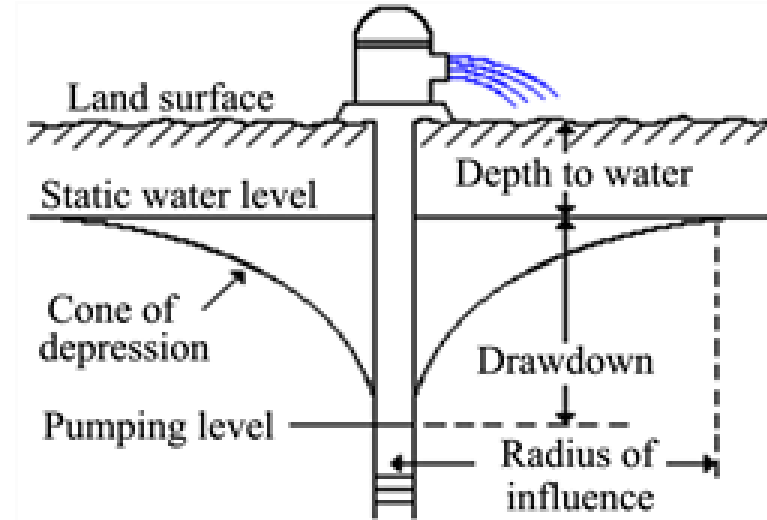


Fig: Diagrammatic section of a pumping well
Source: Raghunath, 1987

Important Terms in Well Hydraulics: Transmissivity

- Transmissivity represents the capacity of an aquifer to transmit water through its full saturated thickness.
- It is defined as the product of hydraulic conductivity (K) and saturated thickness (b): **$T = Kb$**
- Units: L^2/T (e.g., m^2/day)
- Higher T denotes greater groundwater yield and better well productivity.
- For layered aquifers, total transmissivity equals the sum of transmissivity of individual layers.
- Confined aquifer: constant saturated thickness indicating constant T .
- Unconfined aquifer: saturated thickness varies with water table indicating T decreases downgradient.

(Woessner & Poeter, 2020)

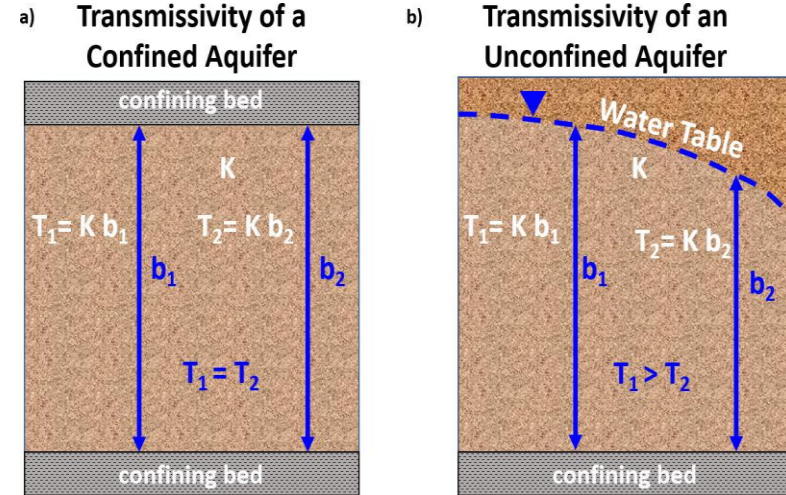


Fig: Transmissivity, T , is the product of hydraulic conductivity, K , and saturated thickness, b , of the aquifer.

Source: Woessner & Poeter, 2020

Important Terms in Well Hydraulics: Storativity

- Storativity (S) is the capacity of an aquifer to store or release water per unit area for a unit change in hydraulic head.
- Also called storage coefficient (older term).
- Dimensionless parameter.
- **Unconfined aquifer:** Storativity is dominated by gravity drainage and is approximately equal to specific yield.
- **Confined aquifer:** Storativity results from compression of the aquifer skeleton and expansion of pore water, giving much smaller values.

(Woessner & Poeter, 2020)

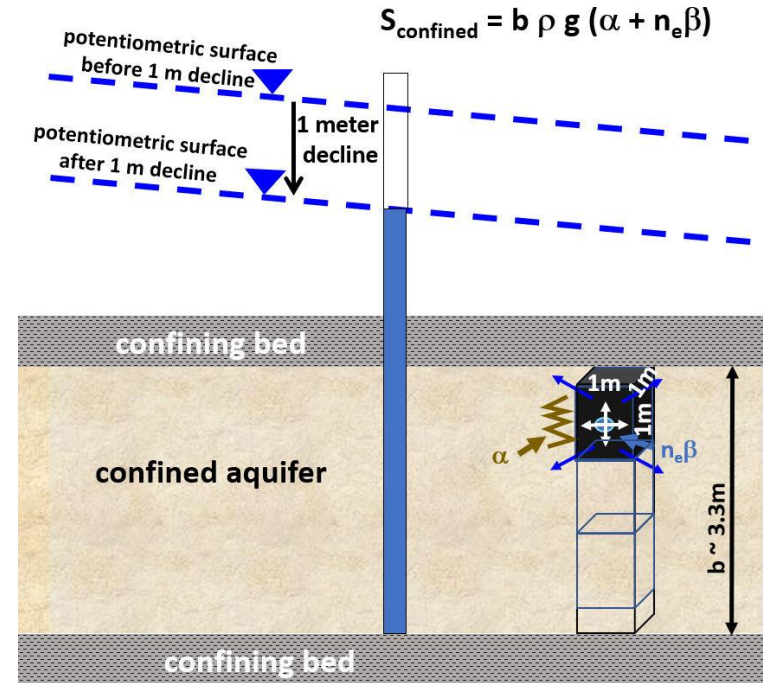


Fig: Parameters controlling storativity, S , of a confined aquifer

Source: Woessner & Poeter, 2020

Important Terms in Well Hydraulics: Yield and Efficiency

Well Yield (Q): The discharge rate at which a well supplies water during pumping. It is usually expressed in L/s or m³/day and depends on aquifer transmissivity, storativity, well design, screen length, and pumping rate.

Well Efficiency (η): A measure of how effectively a well extracts water from an aquifer compared to its ideal or theoretical performance. It reflects losses due to well construction, screen clogging, turbulence, and improper development.

Specific Capacity (Sc): The ratio of pumping rate to the resulting drawdown in the well. It serves as a practical indicator of well performance and aquifer productivity under field conditions.

Specific Yield (Sy): The fraction of water that drains freely under gravity from a saturated unconfined aquifer relative to the total aquifer volume. It governs water release during water-table decline and approximates storativity in unconfined aquifers.

(Testbook, n.d.)

Basic Assumptions in Well Hydraulics

- Aquifer is underlain by a confining layer.
- Geological formations are horizontal and of infinite lateral extent.
- Potentiometric surface is horizontal before pumping.
- Steady initial conditions (no head change before pumping).
- All changes in head are caused only by the pumping well.
- Aquifer is homogeneous and isotropic.
- Groundwater flow is radial toward the well.
- Groundwater flow is horizontal.
- Darcy's law is valid.
- Groundwater has constant density and viscosity.
- Pumping and observation wells are fully penetrating.
- Pumping well has negligible diameter and is 100% efficient.

(Fetter, 2000)

Why the Assumptions?

- To simplify complex groundwater flow systems into mathematically solvable models.
- To allow use of analytical solutions (e.g., Theis, Cooper–Jacob).
- To ensure Darcy's law remains valid under laminar flow conditions.
- To isolate the effect of pumping alone on aquifer response.
- To assume uniform aquifer properties, enabling radial flow analysis.
- To reduce three-dimensional flow into manageable horizontal flow.
- To avoid complications from partial penetration and well losses.
- To make aquifer test interpretation practical and comparable.

Key Takeaways

- Pumping a well **disturbs natural groundwater flow** and creates a hydraulic sink
- Groundwater flows **radially toward wells** due to pressure imbalance
- Drawdown represents the **hydraulic head loss** driving flow to the well
- Cones of depression define the **zone of influence** of pumping
- Aquifer properties (transmissivity, storativity) **control well response**
- Ideal assumptions simplify real systems and form the **basis of well models**

Steady Radial Flow to Non-Leaky Wells

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Introduction to Radial Flow

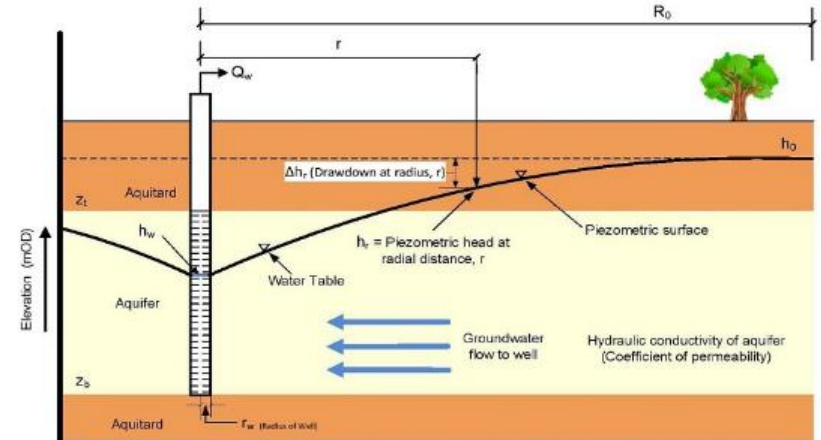
Definition: In groundwater hydraulics, radial flow occurs when water flows towards a pumping well from all directions.

This is a fundamental concept for designing wells and understanding aquifer performance.

Flow Pattern: Water moves horizontally towards center point.

Drawdown: Creates a curved surface known as the "Cone of Depression".

Velocity: Increases as water approaches the well due to decreasing area.



Source: ogi-software.com

"Steady State" Condition



Perfect Equilibrium

The hydraulic system reaches balance. The rate of water inflow (recharge) exactly equals the rate of water outflow (pumping).



Time Independence

Mathematically, the change in head with respect to time is zero. The Cone of Depression stops expanding.



Storage Ignored

The Storage Coefficient (S) is not required. Water levels are stable, so no new water is released from storage.

The Non-Leaky Confined Aquifer

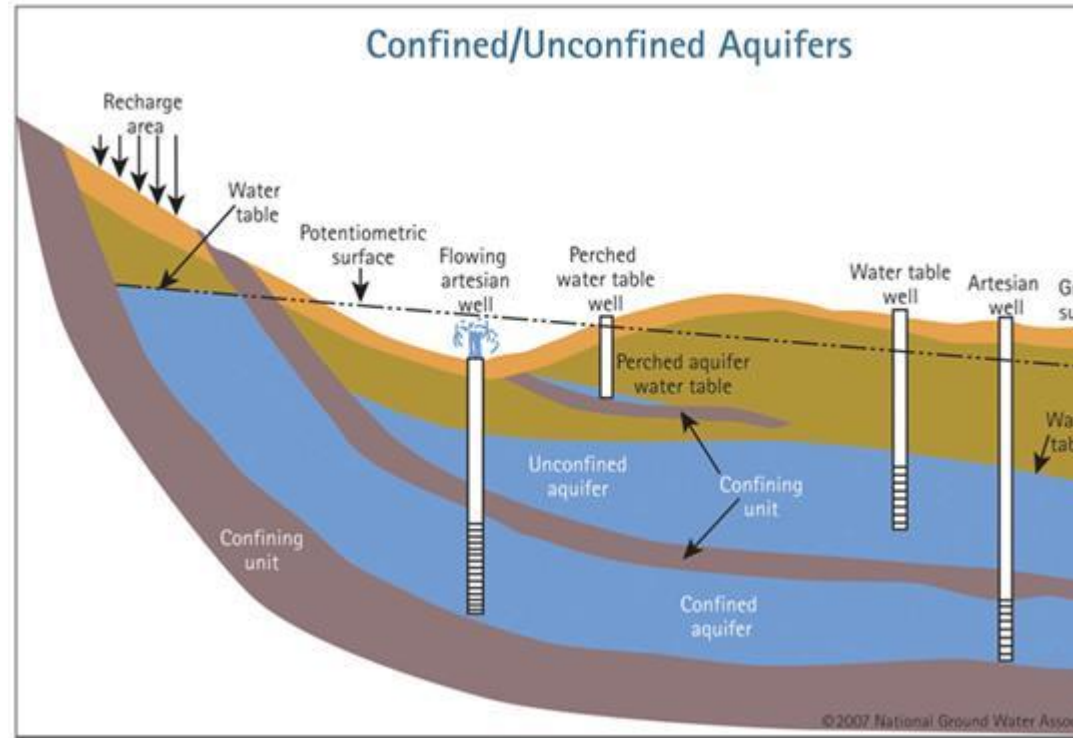
A confined aquifer is a water-bearing layer sandwiched between two impermeable layers (aquicludes).

Key Characteristics:

No Vertical Recharge: Water comes solely from horizontal flow and elastic storage.

Pressure: Water is under pressure; the potentiometric surface sits above the top of the aquifer.

Steady State: Inflow equals Outflow.



<https://www.ngwa.org/what-is-groundwater/About-groundwater/unconfined-or-water-table-aquifers>

Critical Assumptions (Thiem)

Homogeneous & Isotropic: The aquifer material and hydraulic properties are the same everywhere and in all directions.

Infinite Extent: The aquifer extends infinitely in all directions with no boundaries.

Laminar Flow: Darcy's Law applies (no turbulence). Flow is strictly horizontal and radial.

Full Penetration: The well penetrates the entire thickness of the aquifer.

Non-Leaky: Absolutely no recharge from adjacent aquifers above or below.

Steady State: Constant rate of pumping and stabilized drawdown over time.

Governing Equation: Darcy's Law

The Formula

Adapted for cylindrical coordinates, assuming radial flow to a well. Assumes two-dimensional flow to a well centered on a circular island.

$$Q = KA \frac{dh}{dr}$$

Where A is the surface area of a cylinder:

$$A = 2\pi r b$$

Variables

- **Q:** Pumping rate (Constant)
- **K:** Hydraulic Conductivity
- **A:** Area of the cylinder surface (increases with distance r)
- **b:** Aquifer thickness
- **dh/dr:** Hydraulic gradient (slope of the water table)

Derivation: Integration

1. Setup Integral

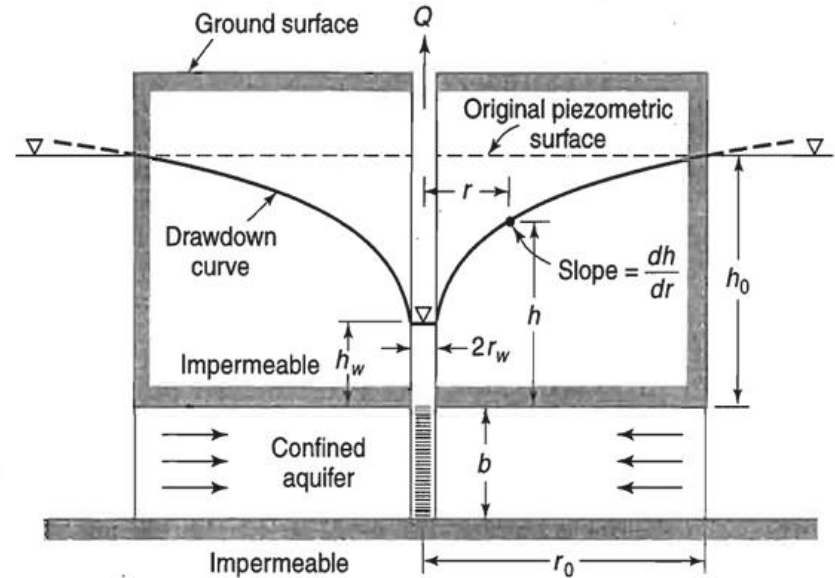
Substitute Area (A) and separate variables (r and h):

$$\frac{dr}{r} = \frac{2\pi K b}{Q} dh$$

2. Integrate Limits

Integrating for the boundary conditions at the well, $h=h_w$ and $r=r_w$ and at the edge of the island, $h=h_o$ and $r=r_o$, we get,

$$\int_{r_w}^{r_o} \frac{dr}{r} = \frac{2\pi K b}{Q} \int_{h_w}^{h_o} dh$$



Todd and Mays, 2005

Derivation: Integration

Rearranging for discharge (Q), we get the standard

Thiem Equation:

The Integrated (Solved) Form

$$\ln \left(\frac{r_o}{r_w} \right) = \frac{2\pi K b}{Q} (h_o - h_w)$$

$$Q = \frac{2\pi K b (h_o - h_w)}{\ln(r_o/r_w)}$$

If h_1 and h_2 are the heads of observation well at distances r_1 and r_2 respectively,

$$Q = \frac{2\pi K b (h_2 - h_1)}{\ln(r_2/r_1)}$$

The Thiem Equation

Rearranging the integral yields the fundamental equation for confined aquifers:

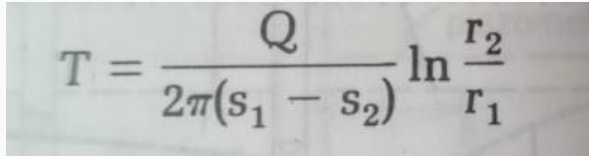
$$Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)}$$

Key Components

- **T (Transmissivity):** $T = K \times b$
- **Unit:** m^2/day
- **$h_2 - h_1$:** Drawdown difference between two observation wells.
- **r_2/r_1 :** Ratio of distances from the pumping well.

The Thiem Equation

From a practical standpoint, the drawdown is measured rather than the head h

A photograph of a handwritten equation on a piece of paper. The equation is the Thiem equation for drawdown in a confined aquifer. It is written in black ink and shows the relationship between discharge rate Q, transmissivity T, drawdowns s1 and s2, and radii r1 and r2.
$$T = \frac{Q}{2\pi(s_1 - s_2)} \ln \frac{r_2}{r_1}$$

The two observation wells at r_1 and r_2 should be close to the pumping well so that drawdown are appreciable and can be readily measured.

Note:

$$\begin{aligned} h_2 - h_1 &= (H_0 - s_2) - (H_0 - s_1) \\ &= H_0 - s_2 - H_0 + s_1 \\ &= (H_0 - H_0) + (s_1 - s_2) \\ &= 0 + (s_1 - s_2) \\ &= s_1 - s_2 \end{aligned}$$

Radius of Influence (R)

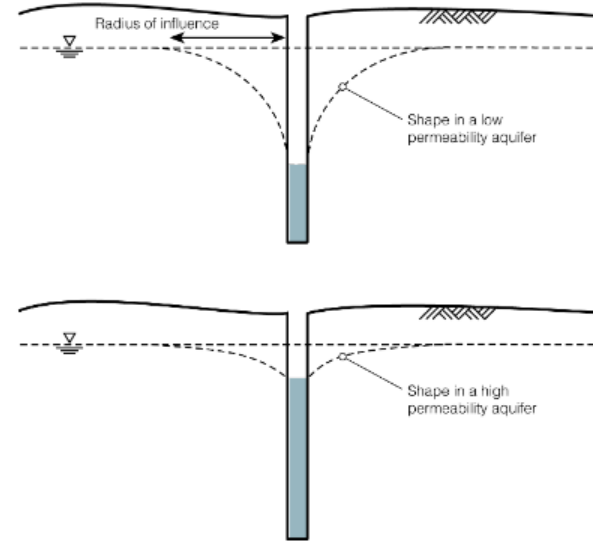
Definition: The radial distance from the well at which drawdown becomes negligible (effectively zero).

$$Q = \frac{2\pi T s_w}{\ln(R/r_w)}$$

Typical Range: commonly 1160 to 500 feet for most construction dewatering wells but can exceed 1,000 feet in very permeable soil or aquifers.

Factors: Affected by pumping rate, transmissivity, and pumping duration.

Note: Confined aquifers typically have a larger Radius of Influence.



Source: repository.lboro.ac.uk

Aquifer Testing (Hydrogeology)

Before any major construction or well drilling begins, scientists perform "pumping tests" to understand the ground itself.

- **Characterization:** By pumping water and observing the drawdown, they can "back-calculate" to find the hidden properties of the soil/rock, specifically **Transmissivity (T)** and **Hydraulic Conductivity (K)**.
- **Environmental Modeling:** These values are essential for computer models used to predict how contaminants (like pollution spills) might travel through the groundwater system.

Using two wells eliminates the need to estimate the uncertain Radius of Influence (R).

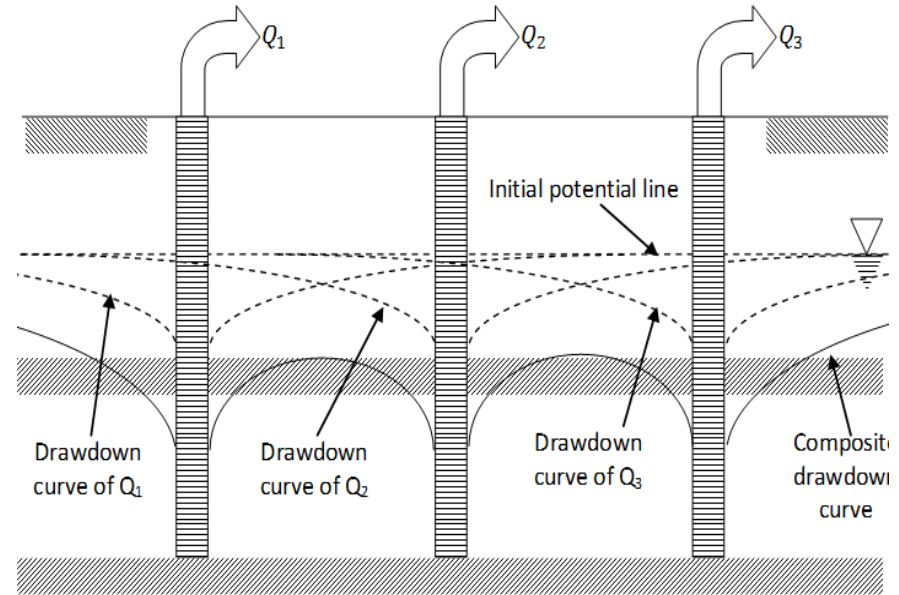


<https://www.stuartwells.co.uk/services/testing-services/pumping-tests/>

Water Supply Engineering

To design efficient water extraction systems for cities or agriculture.

- **Well Spacing:** A crucial part of the math is calculating the "Radius of Influence." If wells are placed too close together, their cones of depression will overlap (interference), significantly reducing the efficiency of both wells.
- **Sustainable Yield:** Estimating how much water can be pumped continuously without depleting the aquifer long-term.



<https://archive.nptel.ac.in/content/storage2/course/s/105103026/module3/lec21/1.html>

Construction Dewatering



<https://www.sulzer.com/en/shared/applications/dewatering-methods>

The Challenge: Deep excavations often require lowering the water table below foundation levels to ensure safety and stability.

The Solution: Engineers use radial flow equations to design "Well Point" systems.

Goal: Calculate the necessary pumping rate (Q) and well spacing to achieve a specific drawdown (s) across the entire construction site.

Limitations & Deviations

Assumption Violations: Real aquifers are rarely perfectly homogeneous or isotropic.

Time to Steady State: Reaching true equilibrium may take days or weeks, which is often impractical.

Partial Penetration: Theory assumes full penetration; partial penetration distorts flow patterns.

Boundary Effects: Real aquifers have physical boundaries (rivers, faults) unlike the "infinite" assumption.

Well Losses: Turbulence near the well screen causes additional drawdown not accounted for in Darcy's Law.

EXAMPLE 4.2.1

A well fully penetrates a 25-m thick confined aquifer. After a long period of pumping at a constant rate of $0.05 \text{ m}^3/\text{s}$, the drawdowns at distances of 50 and 150 m from the well were observed to be 3 and 1.2 m, respectively. Determine the hydraulic conductivity and the transmissivity. What type of unconsolidated deposit would you expect this to be?

SOLUTION

Use Equation 4.2.5 to compute the hydraulic conductivity with $Q = 0.05 \text{ m}^3/\text{s}$, $r_1 = 50 \text{ m}$, $r_2 = 150 \text{ m}$, $s_1 = h_0 - h_1$, and $s_2 = h_0 - h_2$, so $s_1 - s_2 = h_2 - h_1 = 3 - 1.2 = 1.8 \text{ m}$. $Q = 0.05 \text{ m}^3/\text{s} = 4320 \text{ m}^3/\text{day}$, and

$$K = \frac{Q}{2\pi b(h_2 - h_1)} \ln\left(\frac{r_2}{r_1}\right) = \frac{4320 \text{ m}^3/\text{day}}{2\pi(25 \text{ m})(1.8 \text{ m})} \ln\left(\frac{150}{50}\right) = 16.8 \text{ m/day}$$

The transmissivity is $T = Kb = (16.8 \text{ m/day})(25 \text{ m}) = 420 \text{ m}^2/\text{day}$. Referring to Figure 3.2.2 and Table 3.2.1 with $K = 1.94 \times 10^{-4} \text{ m/s}$ shows that this aquifer is probably a medium clean sand. ■

EXAMPLE 4.2.2

A 1-m diameter well penetrates vertically through a confined aquifer 30 m thick. When the well is pumped at 113 m³/hr, the drawdown in a well 15 m away is 1.8 m; in another well 50 m away, it is 0.5 m. What is the approximate head in the pumped well for steady-state conditions and what is the approximate drawdown in the well? Also compute the transmissivity of the aquifer and the radius of influence of the pumping well. Take the initial piezometric level as 40 m above the datum.

SOLUTION

First determine the hydraulic conductivity using Equation 4.2.5: $Q = 113 \text{ m}^3/\text{hr} = 2712 \text{ m}^3/\text{day}$. Then

$$K = \frac{Q}{2\pi b(s_1 - s_2)} \ln\left(\frac{r_2}{r_1}\right) = \frac{2712 \text{ m}^3/\text{day}}{2\pi(30 \text{ m})(1.8 \text{ m} - 0.5 \text{ m})} \ln\left(\frac{50}{15}\right) = 13.3 \text{ m/day}$$

The transmissivity is $T = Kb = 13.3 \text{ m/day} \times 30 \text{ m} = 400 \text{ m}^2/\text{day}$.

To compute the approximate head, h_w , in the pumped well, rearrange Equation 4.2.5 and use $h_2 = h_0 - s_2 = 40 - 0.5 = 39.5 \text{ m}$

$$h_w = h_2 - \frac{Q}{2\pi Kb} \ln\left(\frac{r_2}{r_w}\right) = 39.5 \text{ m} - \frac{2712 \text{ m}^3/\text{day}}{2\pi(13.3 \text{ m/day})(30 \text{ m})} \ln\left(\frac{50 \text{ m}}{0.5 \text{ m}}\right) = 34.5 \text{ m}$$

Drawdown is then

$$s_w = h_0 - h_w = 40 \text{ m} - 34.5 \text{ m} = 5.5 \text{ m}$$

The radius of influence (R) of pumping well can be found by rearranging Equation 4.2.5 and solving for r_0 which is R :

$$R = (r_1) \exp\left[\frac{2\pi Kb(h_0 - h_1)}{Q}\right] = (15 \text{ m}) \exp\left[\frac{2\pi(13.3 \text{ m/day})(30 \text{ m})(40 \text{ m} - 38.2 \text{ m})}{2712 \text{ m}^3/\text{day}}\right] = 79 \text{ m}$$

Unsteady Radial Flow to Non-Leaky Wells

Albert Dhakal(078BCE019)

Unsteady Radial Flow in Wells

- Pumping from a well removes water from **aquifer storage**.
- Area of influence expands outward with time.
- Rate of decline of head * storage coefficient over the area of influence = Q .
- The discharge Q is supplied by the release of water from storage due to a decline in hydraulic head over the area of influence.
- Hydraulic head continues to decline as pumping continues
- Aquifer behaves as **effectively infinite**
- Therefore, flow conditions are **time-dependent (unsteady)**

Drawdown depends on:

- Pumping rate (Q)
- Time since pumping began (t)
- Distance from pumping well (r)
- Aquifer properties (T and S)

Drawdown increases with time, but the **rate of increase gradually decreases**

Assumptions for Unsteady Radial Flow

- Aquifer is homogeneous and isotropic
- Confined aquifer of uniform thickness
- Infinite areal extent
- Well fully penetrates the aquifer
- Flow is horizontal toward the well
- Pumping rate is constant
- Water released instantaneously from storage

Governing Differential Equation

Unsteady radial flow in a confined aquifer is governed by:

$$\partial^2 h / \partial r^2 + (1/r)(\partial h / \partial r) = (S/T)(\partial h / \partial t) \dots 1$$

Where:

- h = hydraulic head
- r = radial distance
- t = time
- T = transmissivity
- S = storage coefficient

- Left side represents spatial variation of head
- Right side represents time variation of head
- Equation is analogous to heat diffusion equation
- Shows groundwater flow behaves as a diffusion process

Physical Basis of the Theis Solution

- Developed by **Theis (1935)**
- Based on analogy between Groundwater flow and Heat conduction
- Pumping well treated as a **mathematical sink of constant strength.**
- Boundary conditions:
 - Initial head uniform ($h=h_0$ for $t=0$)
 - Head unchanged at infinite distance ($h \rightarrow h_0$ for $r \rightarrow \infty$)

$$s = \frac{Q}{4\pi T} \int_u^\infty \frac{e^{-u}}{u} du = \frac{Q}{4\pi T} W(u)$$
$$= \frac{Q}{4\pi T} \left[-0.5772 - \ln u + u - \frac{u^2}{2 \cdot 2!} + \frac{u^3}{3 \cdot 3!} - \frac{u^4}{4 \cdot 4!} + \dots \right] \dots 2$$

where s is drawdown and Q is constant well discharge

(Theis, 1935)

The Well Function

$$u = r^2 S / (4 T t)$$

....3

- u is a **dimensionless time parameter**
- Combines effects of:
 - Distance from well
 - Storage coefficient
 - Time since pumping began
- Controls the shape of drawdown curve

- $W(u)$ is an **exponential integral**
- Expressed as a convergent series:

$$W(u) = -0.5772 - \ln u + u - \frac{u^2}{2 \cdot 2!} + \frac{u^3}{3 \cdot 3!} - \dots$$

- Represents **time-dependent response** of aquifer
- Tabulated or plotted as **type curves**

Applicability of the Nonequilibrium Equation

- The assumptions of the nonequilibrium (Theis) equation are **rarely fully satisfied in the field**
- Understanding these assumptions helps recognize the **approximations involved**
- Average values of **transmissivity (T)** and **storage coefficient (S)** can be obtained from:
 - Drawdown measurements in one or more observation wells
 - Constant-rate pumping tests
- Due to mathematical complexity of the Theis equation, **simpler approximate methods** have been developed
- Commonly used methods include: **Theis method, Cooper–Jacob method, Chow method**
- Each method uses supporting **tables or graphs**.

Theis Method

- Simplified form: $s = (Q / 4\pi T) W(u)$; where $W(u)$ is the well function
- Also; $r^2 / (4Tt) = u/s$
- $W(u)$ vs u plotted on log–log paper (type curve)
- Observed drawdown data superimposed on type curve.
- Parallel shifting until curves match
- Matching point gives T and S
- Table provides $W(u)$ for a wide range of u values
- Used for graphical estimation of aquifer parameters and standard reference in pumping test analysis

Table 4.4.1 Values of $W(u)$ for Values of u

u	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
$\times 1$	0.219	0.049	0.013	0.0038	0.0011	0.00036	0.00012	0.000038	0.000012
$\times 10^{-1}$	1.82	1.22	0.91	0.70	0.56	0.45	0.37	0.31	0.26
$\times 10^{-2}$	4.04	3.35	2.96	2.68	2.47	2.30	2.15	2.03	1.92
$\times 10^{-3}$	6.33	5.64	5.23	4.95	4.73	4.54	4.39	4.26	4.14
$\times 10^{-4}$	8.63	7.94	7.53	7.25	7.02	6.84	6.69	6.55	6.44
$\times 10^{-5}$	10.94	10.24	9.84	9.55	9.33	9.14	8.99	8.86	8.74
$\times 10^{-6}$	13.24	12.55	12.14	11.85	11.63	11.45	11.29	11.16	11.04
$\times 10^{-7}$	15.54	14.85	14.44	14.15	13.93	13.75	13.60	13.46	13.34
$\times 10^{-8}$	17.84	17.15	16.74	16.46	16.23	16.05	15.90	15.76	15.65
$\times 10^{-9}$	20.15	19.45	19.05	18.76	18.54	18.35	18.20	18.07	17.95
$\times 10^{-10}$	22.45	21.76	21.35	21.06	20.84	20.66	20.50	20.37	20.25
$\times 10^{-11}$	24.75	24.06	23.65	23.36	23.14	22.96	22.81	22.67	22.55
$\times 10^{-12}$	27.05	26.36	25.96	25.67	25.44	25.26	25.11	24.97	24.86
$\times 10^{-13}$	29.36	28.66	28.26	27.97	27.75	27.56	27.41	27.28	27.16
$\times 10^{-14}$	31.66	30.97	30.56	30.27	30.05	29.87	29.71	29.58	29.46
$\times 10^{-15}$	33.96	33.27	32.86	32.58	32.35	32.17	32.02	31.88	31.76

Table: Values of $W(u)$ for u
Source: Todd and Mays, 2005

Cooper-Jacob Method

- For **small r** and **large t**, higher-order terms of the Theis well function become negligible.
- Drawdown varies linearly with **logarithm of time**.
- The slope of this line is used to compute **transmissivity**, while the time intercept is used to estimate **storage coefficient**.

$$s = \frac{Q}{4\pi T} \left(-0.5772 - \ln \frac{r^2 S}{4Tt} \right)$$

Rewriting and changing to decimal logarithms reduce this to

$$s = \frac{2.30Q}{4\pi T} \log \frac{2.25Tt}{r^2 S}$$

Therefore, a plot of drawdown s versus the logarithm of t forms a straight line. Projecting this line to $s = 0$, where $t = t_0$ (see Figure 4.4.2), we have

$$0 = \frac{2.30Q}{4\pi T} \log \frac{2.25Tt_0}{r^2 S}$$

and it follows that for the above to hold true, $\log(1) = 0$ and

$$\frac{2.25Tt_0}{r^2 S} = 1$$

resulting in

$$S = \frac{2.25Tt_0}{r^2}$$

A value for T can be obtained by noting that if $t/t_0 = 10$, then $\log t/t_0 = 1$; therefore, replacing s by Δs , where Δs is the drawdown difference per log cycle of t , Equation 4.4.8 becomes

$$T = \frac{2.30Q}{4\pi \Delta s}$$

Cooper-Jacob method for solution of the non-equilibrium equation

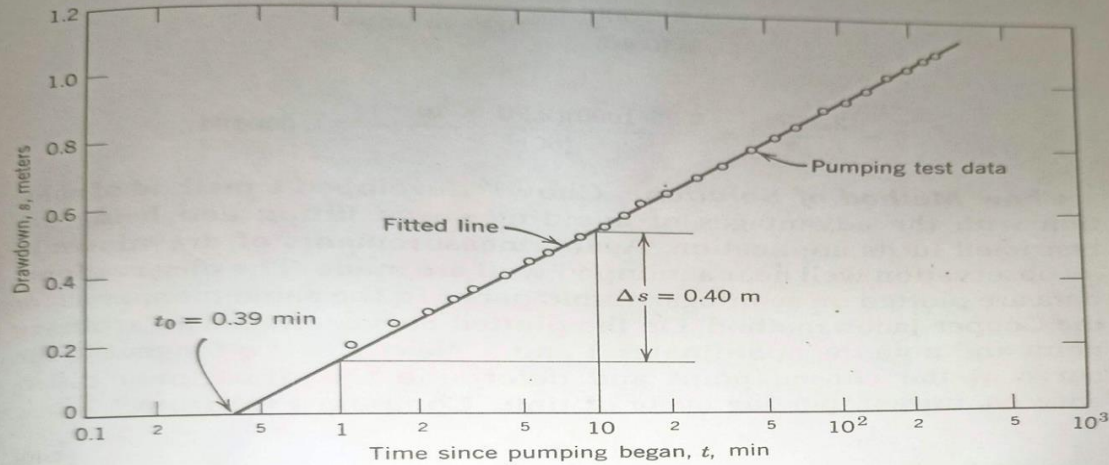


Fig. 4.10 Cooper-Jacob method for solution of the nonequilibrium equation.

Chow Method

Purpose

- Alternative graphical method
- Avoids curve fitting
- Works directly on semilog plot

Procedure

1. Plot **s vs log(t)** (same as Cooper–Jacob)
2. Select any point on the curve
3. Draw a **tangent** at that point
4. Measure drawdown difference Δs per log cycle
5. From $F(u)$ find $W(u)$ and u .
$$F(u) = \frac{s}{\Delta s}$$

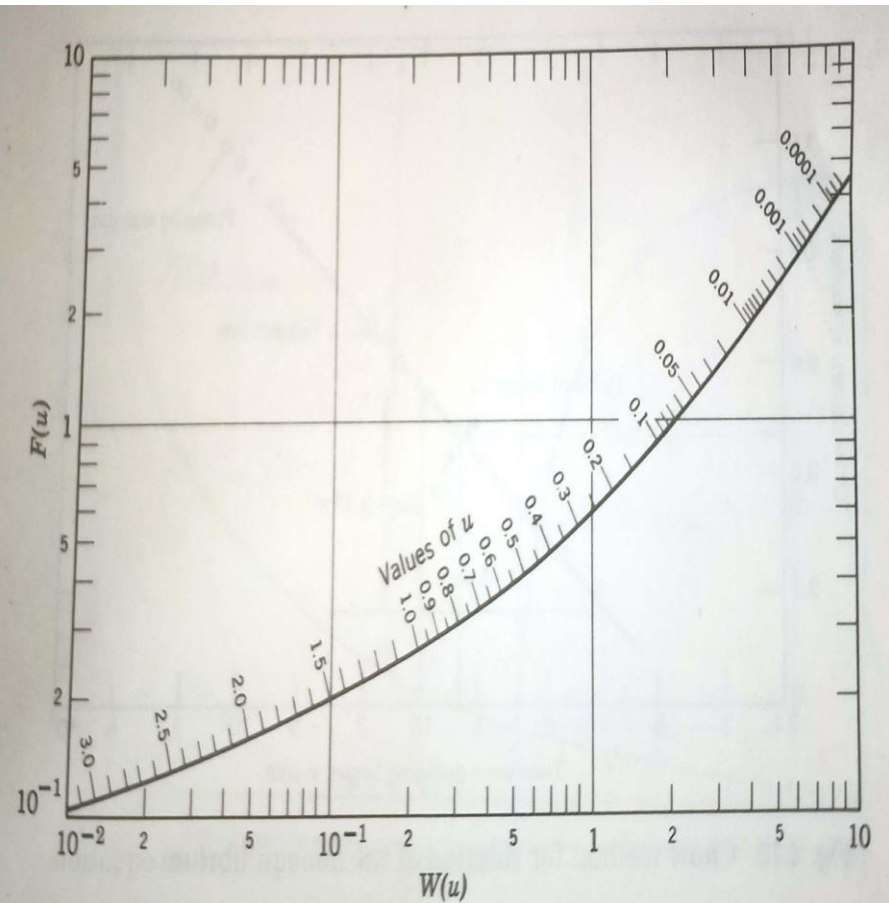
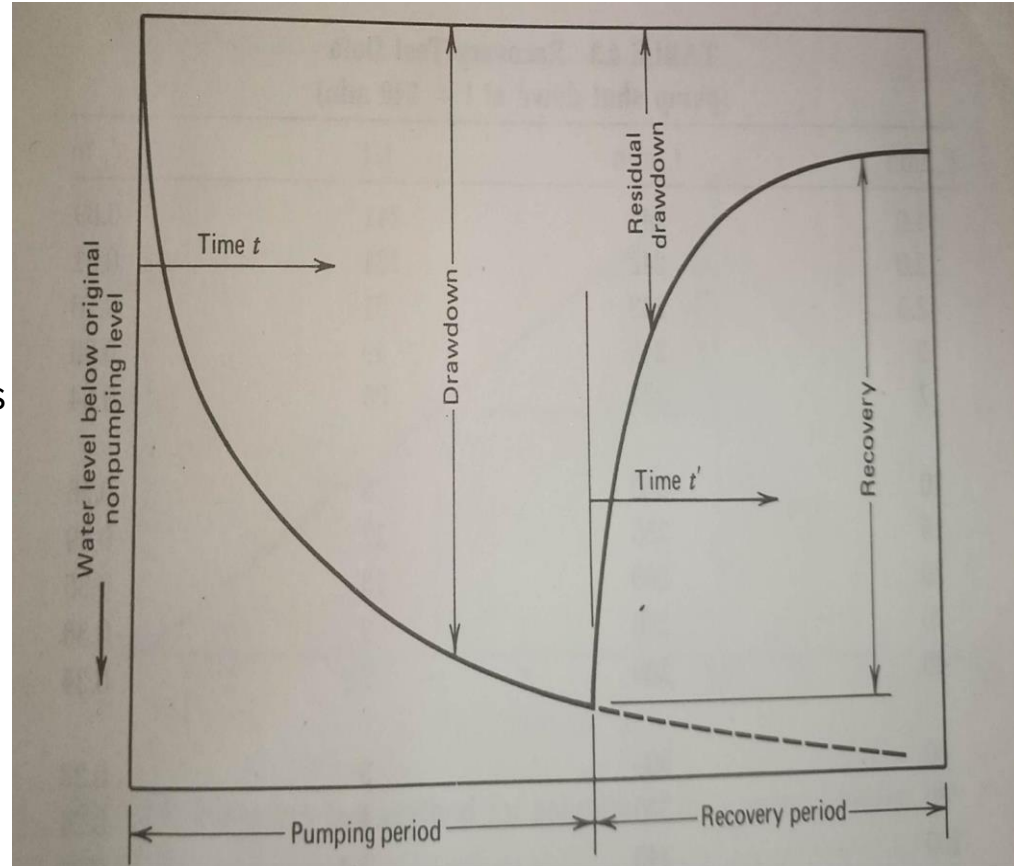


Fig. 4.11 Relation among $F(u)$, $W(u)$, and u (After Chow¹⁰).

Recovery Test

- After pumping stops, water levels rise; the remaining drawdown is called **residual drawdown (s')**.
- Plot **s' versus $\log(t / t')$** ; the plot forms a **straight line** at late time.
- **Transmissivity (T)** is obtained from the slope of the line.
- **Storage coefficient (S) cannot be determined** from the recovery test.



Partially Penetrating Wells

Ashesh Chandra Bhattarai(078BCE029)

Partially Penetrating Wells

- A well whose length of water entry is less than the aquifer it penetrates is known as a **partially penetrating well**.
- Many artesian wells do not penetrate the full thickness of the aquifer and many wells are provided with strainers with only a portion of thickness of the aquifer.
- In such cases, there is a **vertical convergence** of streamlines near the well, resulting increased **entrance velocity** with increased **drawdown**.

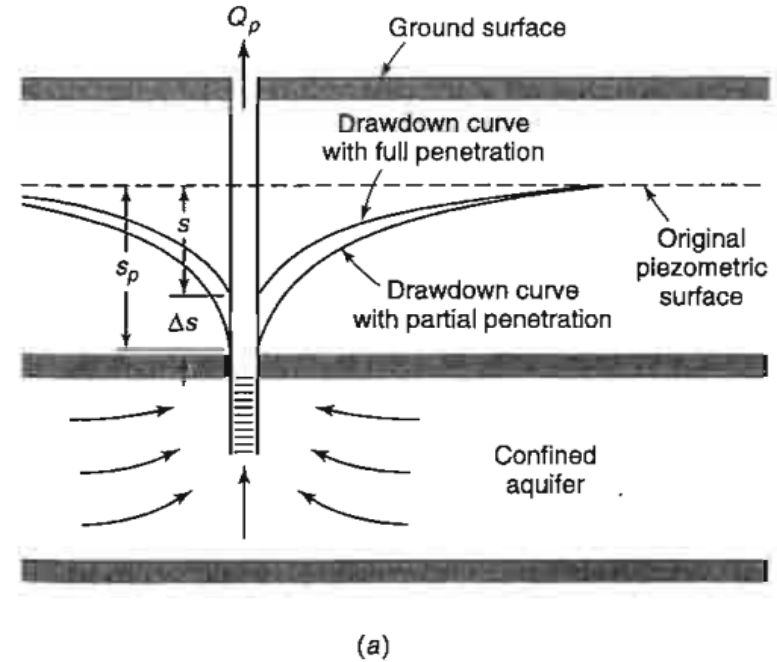


Fig: Effect of partially penetrating well on drawdown
Source: Todd, D. K., & Mays, L. W. (2005).
Groundwater hydrology (3rd ed.). John Wiley & Sons.

Partially Penetrating Wells

- The average length of a flow line into a partially penetrating well exceeds that into a fully penetrating well, and hence a greater resistance to flow is encountered.
- Relationship between two similar wells, one partially and the other fully penetrating the same aquifer:

If $Q_p = Q$, then $s_p > s$

If $s_p = s$, then $Q_p < Q$

- The effect of partial penetration is negligible on the flow pattern and the drawdown beyond a radial distance larger than 0.5 to 2 times the saturated thickness 'b' , depending on the amount of penetration.

Partially Penetrating Wells

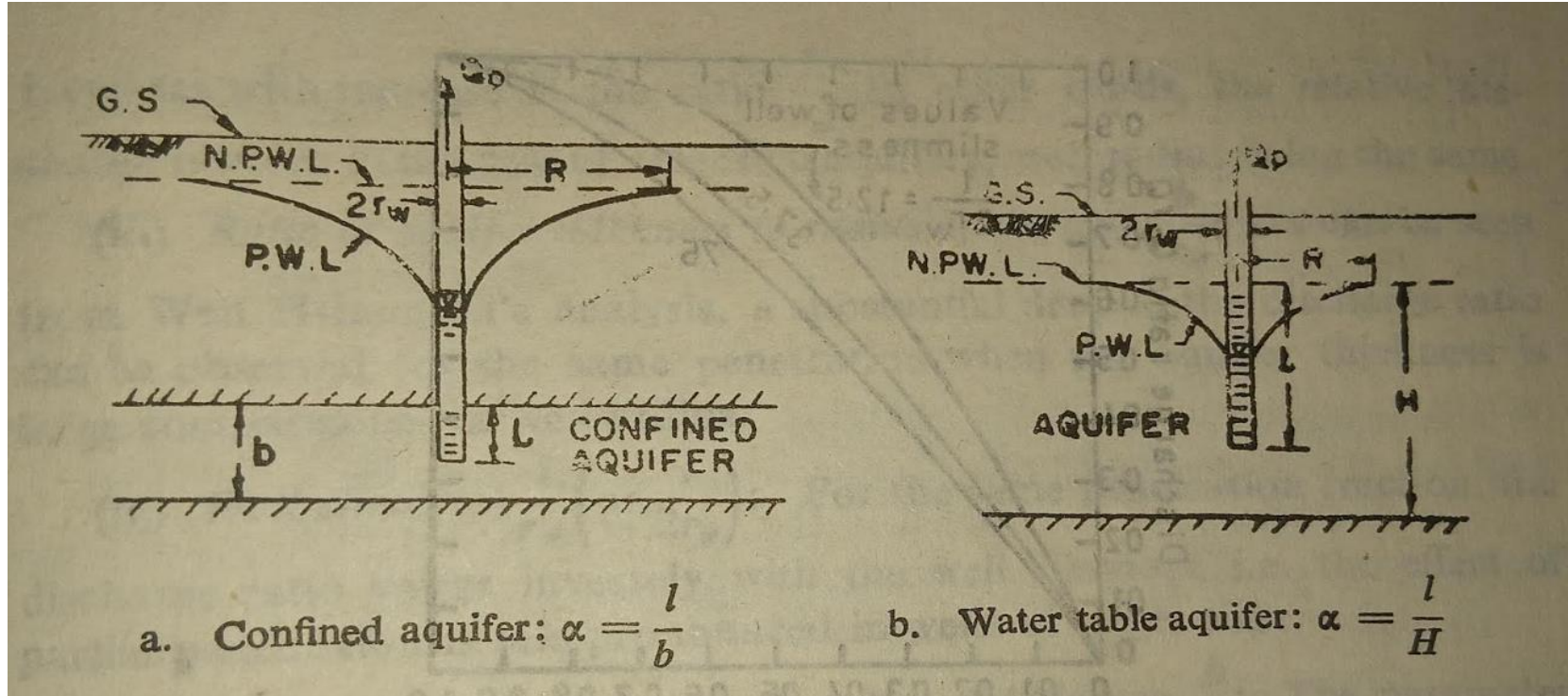


Fig: Partial penetration of wells

Source: Raghunath, H. M. (1987). Ground water (2nd ed.). Wiley Eastern Limited.

Partially Penetrating Wells

Kozeny's formula for a partially penetrating artesian well

$$Q_p = Q \left[\alpha \left(1 + 7 \sqrt{\frac{r_w}{2\alpha b}} \cos\left(\frac{\pi\alpha}{2}\right) \right) \right] \dots\dots\dots(1)$$

Where:

- Q_p = discharge of partially penetrating well
- Q = discharge of fully penetrating artesian well
- $\alpha = \frac{l}{b}$ = penetration ratio
- b = thickness of confined aquifer
- r_w = radius of well

Partially Penetrating Wells

- Based on the data obtained from a series of tests on an **electric analog model**, Wein Hsiung Li (1954) proposed a simple empirical formula for the ratios of drawdown and discharge in a partially penetrating well.

$$\frac{S_{wp}}{S_w} = 1 + \frac{\log \frac{b}{r_w}}{\log \frac{R}{r_w}} \left[\left(\frac{1}{\alpha} \right)^{0.75} - 1 \right] \dots\dots\dots(2)$$

$$\frac{Q_p}{Q} = \frac{1}{1 + \frac{\log \frac{b}{r_w}}{\log \frac{R}{r_w}} \left[\left(\frac{1}{\alpha} \right)^{0.75} - 1 \right]} \dots\dots\dots(3)$$

Here, s_{wp} = drawdown in a partially penetrating well

s = drawdown in a fully penetrating well

b = aquifer thickness

r_w = radius of the well

R = radius of influence

α = penetration ratio

These formulae are applicable for $\alpha > 0.1$.

Partially Penetrating Wells

Q) A **40 cm diameter well** penetrates an aquifer of **30 m thickness** and the length of the screened portion is **10 m**. The well yields **2000 litres per minute** with a **drawdown of 3 m** in the well. If the length of the screen is increased to **20 m**, determine: a) The **drawdown in the well**, and b) The **percentage increase in the specific capacity** of the well.

Assume a **homogeneous aquifer**

Solution:

Partially Penetrating Wells

Q) A **40 cm diameter well** penetrates an aquifer of **30 m thickness** and the length of the screened portion is **10 m**. The well yields **2000 litres per minute** with a **drawdown of 3 m** in the well. If the length of the screen is increased to **20 m**, determine: a) The **drawdown in the well**, and b) The **percentage increase in the specific capacity** of the well.

Assume a **homogeneous aquifer**

Solution:

Use of Muskat (1964) equation;

$$\left(\frac{Q}{s_w}\right)_p = \left(\frac{Q}{s_w}\right) \left[\alpha \left(1 + 7 \sqrt{\frac{r_w}{2\alpha b}} \cos\left(\frac{\pi\alpha}{2}\right) \right) \right]$$

L = screen length,

α = (fraction of penetration) = **L/b**

$\alpha_1 = 10/30 = 1/3$ and $\alpha_2 = 20/30 = 2/3$

For same **Q**,

a) $s_{w2} = 1.92$ m

Partially Penetrating Wells

2) Calculation of specific capacity:

$$\text{Specific capacity} = \frac{Q}{s_w}$$

Initial specific capacity

$$\frac{Q}{s_{w1}} = \frac{2}{3} = 0.67 \text{ m}^2/\text{min}$$

New specific capacity

$$\frac{Q}{s_{w2}} = \frac{2}{1.92} = 1.04 \text{ m}^2/\text{min}$$

Increase in specific capacity

$$\% \text{ increase} = \frac{1.04 - 0.67}{0.67} \times 100 = 55\%$$

Partially Penetrating Wells

Confined Aquifers:

- Increase in drawdown is due to the well not fully penetrating the aquifer or only a portion of the aquifer screened, resulting in higher entrance velocities and convergence of streamlines .
- The drawdown in the partially penetrating well, $s_{wp} = s_w + \Delta s_w$
- The disposition of the screen in the aquifer is shown in figure. For steady-state conditions, it can be shown that:

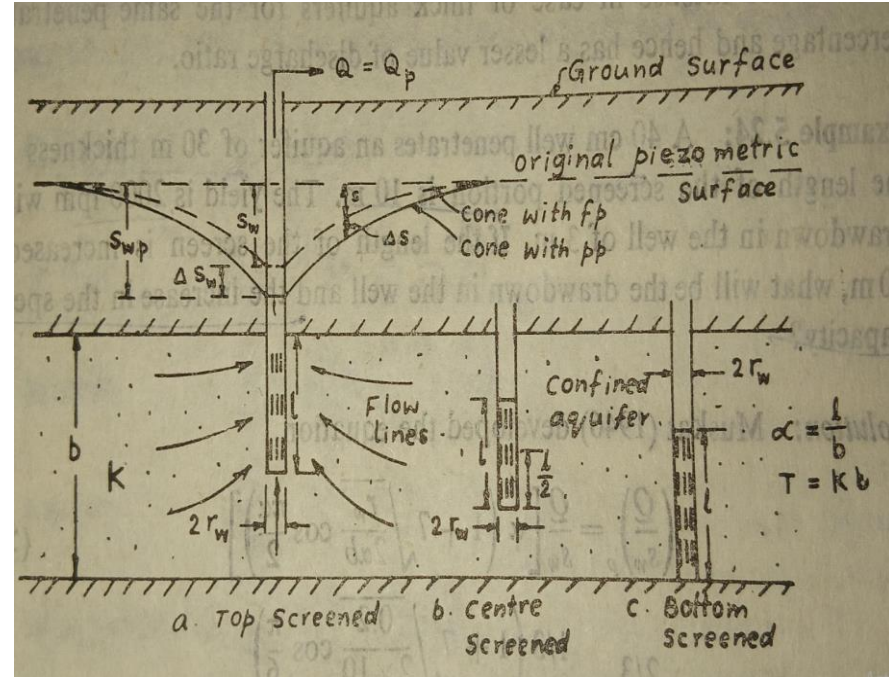


Fig: Partially penetrating well in a confined aquifer
Source:Raghunath, H. M. (1987). Ground water(2nd ed.).
Wiley Eastern Limited.

Partially Penetrating Wells

(i) when the screen is located at the top or bottom of the aquifer,

$$\alpha = \frac{l}{b}, \quad T = Kb$$

$$\Delta s_w = \frac{Q}{2\pi T} \frac{1-\alpha}{\alpha} \ln \left(\frac{1-\alpha}{r_w} l \right), \quad \alpha > 0.2$$

(ii) when the screen is centrally located

$$\Delta s_w = \frac{Q}{2\pi T} \frac{1-\alpha}{\alpha} \ln \left(\frac{1-\alpha}{2r_w} l \right)$$

Unconfined aquifer:

For a fully penetrating well, for steady-state conditions,

$$H^2 - h_w^2 = \frac{Q}{\pi K} \ln \frac{R}{r_w}$$

Partially Penetrating Wells

For a partially penetrating well (usually bottom-half to one-third of the aquifer),

$$\alpha = l/h_w, T = KH, (h_w = H - s_w)_{fp}$$

So,

$$\Delta s_w \cdot 2H = \frac{Q}{\pi K} \frac{1 - \alpha}{\alpha} \ln \left(\frac{1 - \alpha}{r_w} l \right)$$

Then,

$$(H^2 - h_w^2)_{pp} = (H^2 - h_w^2)_{fp} + \Delta s_w \cdot 2H$$

from which,

$$(s_w)_{pp} = H - (h_w)_{pp}$$

where the subscripts:

pp — partial penetration,

fp — full penetration.

Partially Penetrating Wells

Q) A 30 cm well penetrates fully a homogeneous aquifer having a permeability of 40 m/day. The saturated thickness of the aquifer is 15 m. What is the drawdown in the well when pumped at the rate of 1000 lpm? What will be the drawdown if only the bottom half of the aquifer is screened? Assume a radius of influence of 300 m.

Partially Penetrating Wells

Q) A 30 cm well penetrates fully a homogeneous aquifer having a permeability of 40 m/day. The saturated thickness of the aquifer is 15 m. What is the drawdown in the well when pumped at the rate of 1000 lpm? What will be the drawdown if only the bottom half of the aquifer is screened? Assume a radius of influence of 300 m.

(i) Fully penetrating well

$$H^2 - h_w^2 = \frac{2.3Q}{\pi K} \log \frac{R}{r_w}$$

$$15^2 - h_w^2 = \frac{2.3 \times (1/60)}{\pi \times (40/86400)} \log \frac{300}{0.15}$$

$$15^2 - h_w^2 = 86.5$$

Therefore,

$$h_w^2 = 138.5, \quad h_w = 11.76 \text{ m}$$

$$s_w = 15 - 11.76 = 3.24 \text{ m}$$

(ii) Partially penetrating well

Length of well screen = $\frac{1}{2} \times 15 = 7.5$ m from the bottom of the aquifer.

$$\alpha = \frac{l}{h_w} = \frac{7.5}{11.767} \approx 0.637, \quad 1 - \alpha \approx 0.363$$

$$\Delta s_w \cdot 2H = \frac{2.3026 \times (1/60)}{\pi \times (40/86400)} \cdot \frac{0.363}{0.637} \cdot \log \left(\frac{0.363 \times 7.5}{0.15} \right) \approx 18.9$$

$$(H^2 - h_w^2)_{pp} = 86.5 + 18.9 = 105.4$$

$$(15^2 - h_w^2)_{pp} = 105.4$$

$$(h_w)_{pp} \approx 10.94 \text{ m}$$

$$s_{wp} = 15 - 10.94 \approx 4.06 \text{ m}$$

$$\Delta s_w = 4.06 - 3.23 \approx 0.83 \text{ m}$$

Multiple Well Systems and Interference

Amish Khatiwada(078BCE020)

Introduction

- Multiple well systems in well hydraulics involve the analysis of groundwater flow when several wells are pumping from the same aquifer, leading to overlapping cones of depression and well interference

Well Interference

- In the case of some multiple wells, the cones of depression of individual may intersect with each other.
- When the cones of depression of two or more wells overlap, the groundwater table of both wells combine to form a new phreatic line combining the two together.

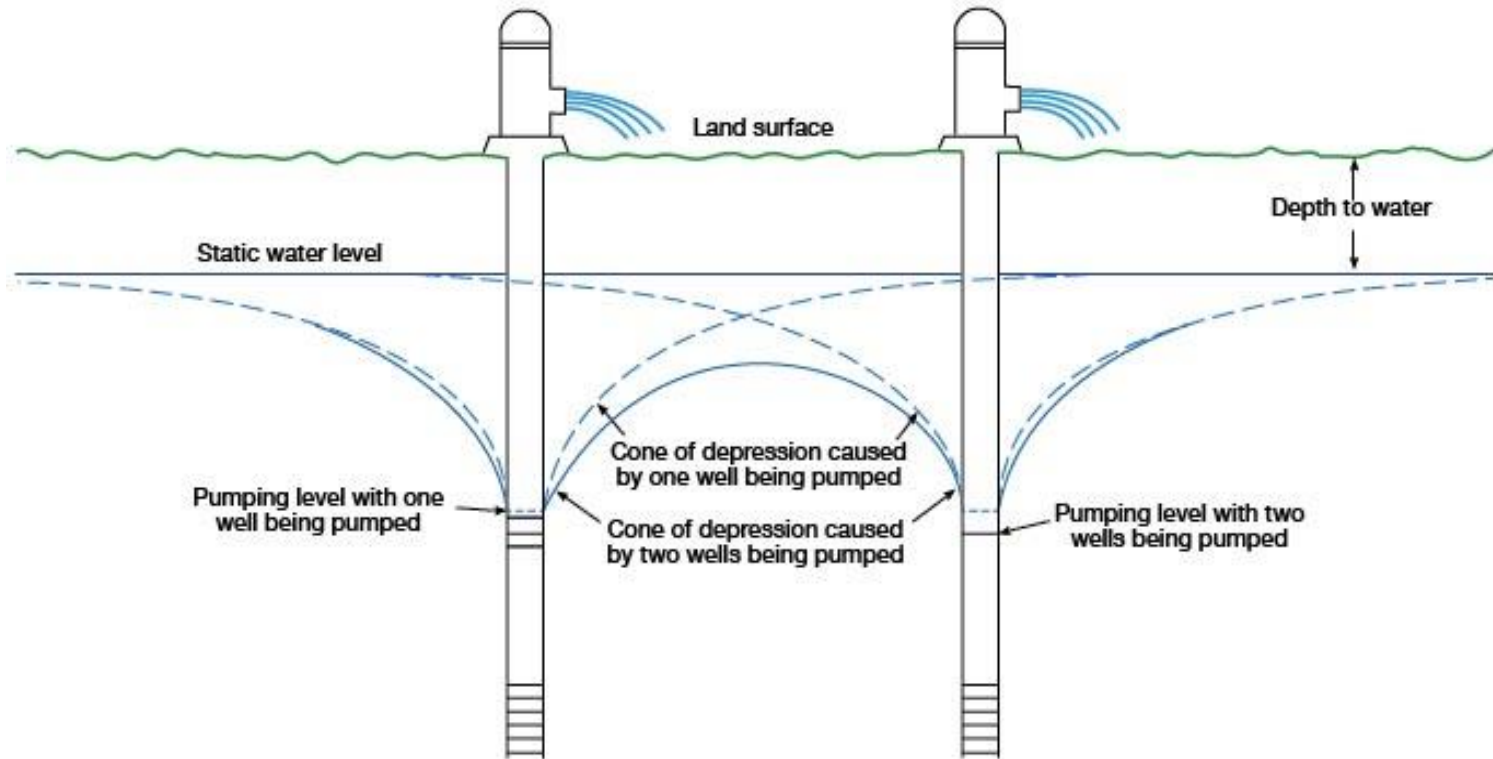


Fig: Combined cone of depression

Effects of Well Interference

- The total groundwater output is less than the sum of the discharging capacity of individual wells.
- The efficiency of each well is decreased.
- The drawdown is increased and as a result pumping lift has become higher.
- The pumping cost has increased due decrease in efficiency and increase in pumping lift.

Flow in Multiple Well Systems(Confined)

- In confined aquifer systems, the principle of superposition applies, allowing the total drawdown at any point to be calculated as the sum of individual drawdowns caused by each pumping well, due to the linearity of the Laplace equation under steady-state conditions.
- This principle is used to determine composite drawdowns in well fields, which is essential for designing efficient water supply systems.

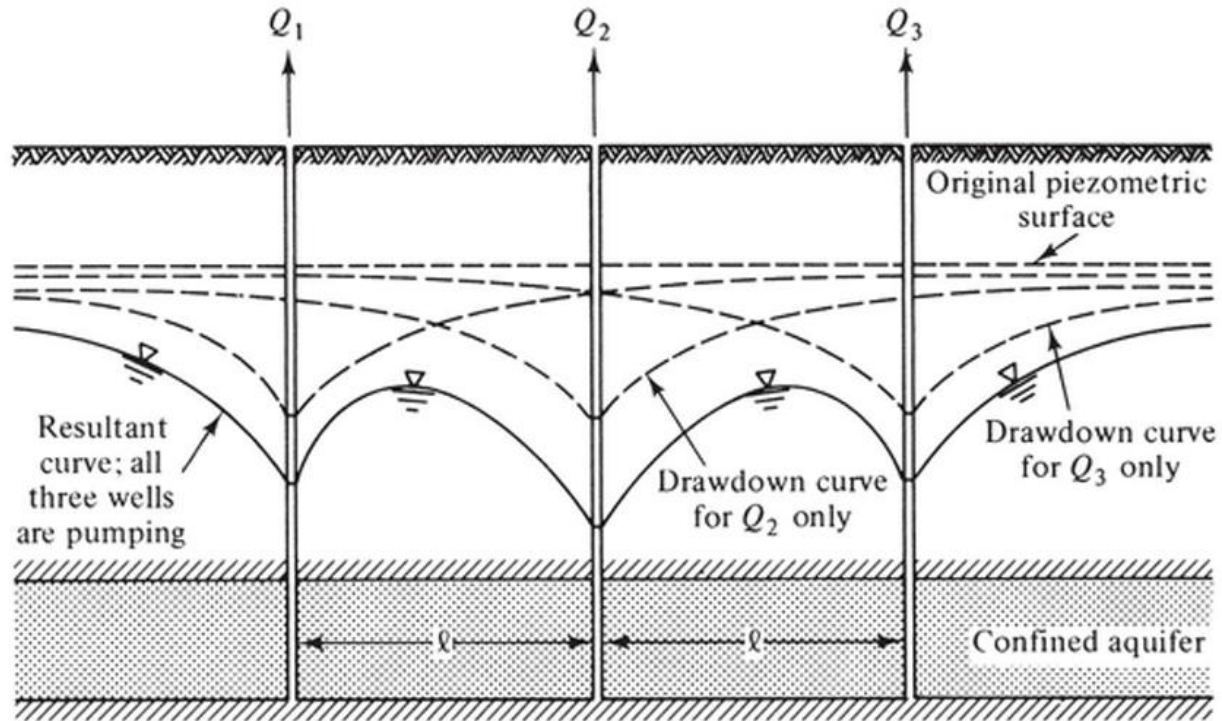


Fig : Individual and composite drawdown curves for three wells on a line.
Source: Sandhyavitri et. al., 2019

Unconfined aquifers

- For unconfined aquifers, the linear superposition principle is not directly applicable because significant drawdown reduces the saturated thickness and thus the aquifer transmissivity, leading to higher composite hydraulic gradients.
- To account for this, a correction procedure is required: first, theoretical confined drawdowns are calculated for each well, then a correction is applied using the initial saturated thickness to estimate the actual unconfined drawdown. This ensures more accurate predictions of drawdown in systems where the water table is significantly lowered.

Uses of Multiple Well Systems

- Multiple Well systems are used for water supply, subsurface drainage, dewatering for excavation or mining, and managing seawater intrusion in coastal aquifers.
- In water supply applications, wells are typically spaced to minimize interference and reduce drawdown, while in drainage or dewatering contexts, wells are arranged closely to maximize interference and lower the water table.
- Configurations such as linear, triangular, and square arrangements of closely spaced wells have been analyzed for their discharge characteristics under steady-state conditions.

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